CHUNKING AND MEMORY VS RECALL

1. Project Title

Memory Recall Task using Chunking Technique with Timer Automation

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2. Objective

The objective of this project is to design and develop an interactive **memory recall application** based on the psychological concept of **chunking**, which helps improve short-term memory retention.

The system displays grouped visual elements (football club logos) for a limited time and evaluates the user's ability to recall them.

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3. Concept Used

🔹 Chunking Concept

Chunking is a memory technique where information is grouped into meaningful units (chunks) to make it easier to remember.

In this project:

- 10 football club logos are grouped together.
- Instead of remembering individual images randomly, users remember them as a logical category (football teams).
- This improves recall performance.

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4. Tools & Platform Used

- **Platform:** Sigma (UI Design & Prototyping)
- **Concept:** Cognitive Psychology – Chunking
- **Logic Used:**
    - Frame transition

- Timer-based auto navigation
- Input validation
- Score calculation

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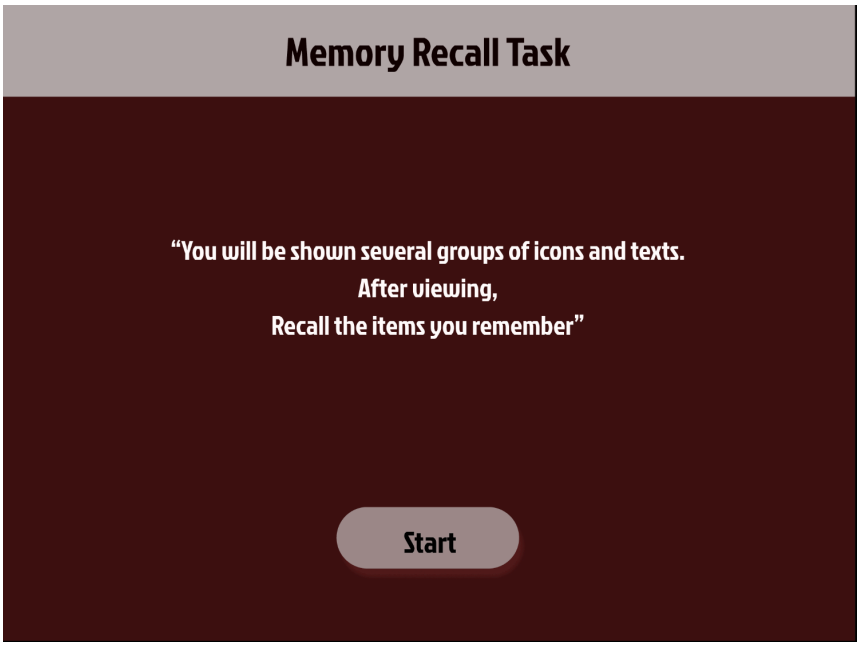
5. System Workflow

Step 1 – Start Screen



- Displays project title: *Memory Recall Task*
- Instructions shown to user.
- Start button initiates the test.

Step 2 – Display Phase (Timed Viewing)

- 10 football club logos are displayed.
- Timer countdown shown (5 → 4 → 3 → 2 → 1 seconds).
- Frames automatically transition using timer logic.
- User cannot interact during this phase.

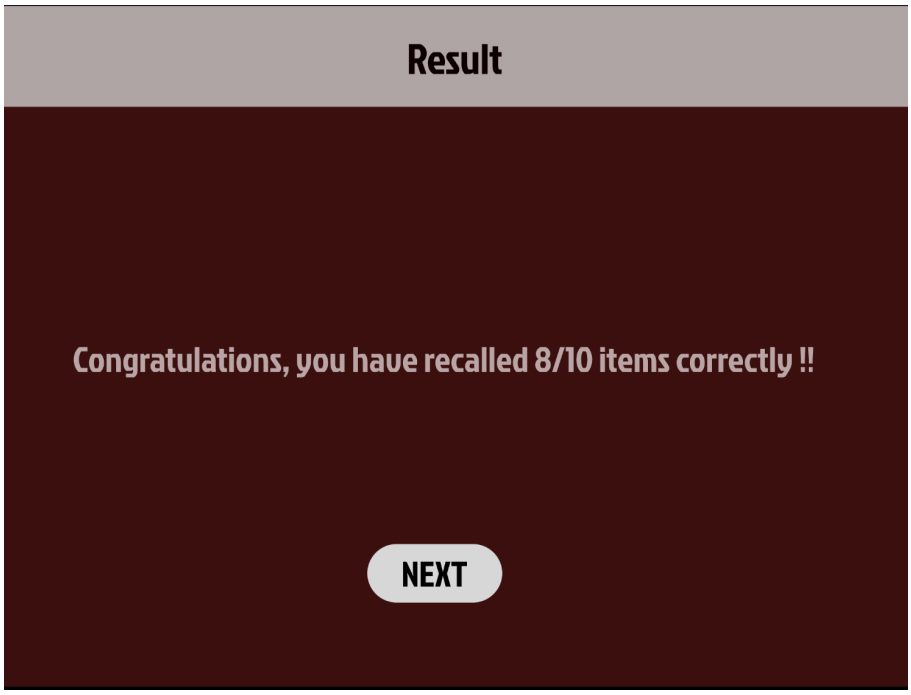
Displayed Logos:

- Barcelona
- Real Madrid
- Bayern Munich
- Manchester United
- Liverpool
- Manchester City
- Chelsea
- Arsenal
- Paris Saint-Germain
- Tottenham Hotspur

Step 3 – Recall Phase

- All logos disappear.
- User is given 10 input fields.
- User types the team names they remember.
- Submit button evaluates the response.

Step 4 – Result Phase



- System checks correct matches.

Displays score in format:
`Congratulations, you have recalled 8/10 items correctly!!`

- NEXT button resets or moves to next activity.

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6. Key Features

✔ Timer-based automatic frame transition
✔ Chunk-based grouped information
✔ Interactive recall input system
✔ Automated score calculation
✔ Clean and minimal UI design
✔ Psychological learning principle applied

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7. Technical Implementation Details

Frame Structure in Sigma

1. Frame 1 – Welcome Screen
2. Frame 2–6 – Timer Countdown Screens
3. Frame 7 – Recall Input Screen
4. Frame 8 – Result Screen

Timer Logic

- Each frame is connected using timed animation.
- Auto-navigation set to 1-second delay per frame.
- After countdown completes → Redirect to Recall Phase.

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8. Scoring Logic

- Each correctly recalled team name = 1 mark
- Total items = 10
- Final score displayed as:
  `Correct Answers / 10`

Matching can be case-insensitive for better user experience.

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9. Learning Outcomes

Through this project, I learned:

- Application of chunking in UI-based memory systems
- Timer-based frame automation in Sigma
- Designing interactive cognitive tasks
- Creating multi-screen UI flows
- Implementing score evaluation logic

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10. Future Enhancements

- Add multiple difficulty levels
- Randomize logo positions
- Add different categories (countries, animals, brands, etc.)
- Add leaderboard system
- Store results in database
- Add sound effects for countdown

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11. Conclusion

This project successfully demonstrates how the psychological principle of **chunking** can be applied in an interactive digital system to improve memory recall.

The timer-based controlled exposure and structured recall phase make it a simple yet effective cognitive training tool.